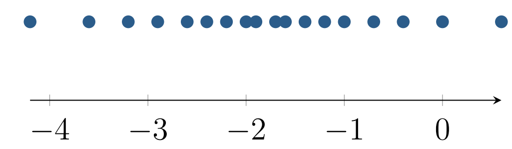
Two Measurements per Technician

Unit 4 · Topic 4.4 · TODAY'S PROBLEM

Suppose a company takes an SRS of 18 of its 300 technicians. Each completes the same task with old and new interfaces; interface order is randomly assigned. The plot shows d = new time − old time. Researchers ask whether the new interface reduces the population mean time.



New-interface time minus old-interface time (seconds)

Nia: "The technician seems to be the independent unit, so I would build one difference per person; the subtraction order matters for the tail."

Colin: "There are 36 recorded times, so I would use two independent samples of 18 and a two-sided alternative."

FIRST TAKE (5 min, on your sheet)

Which setup better aligns with the study? Cite the measurement structure or the direction of the claim.

DOK 1 **(a)** Define μ_d and state hypotheses for the research question.

DOK 2 **(b)** Calculate $\bar{d}$ and s_d, verify random, 10 percent, and shape conditions, and name the procedure.

DOK 3 **(c)** Adjudicate the setups using the same 18 technicians, subtraction order, directional question, and difference plot. Explain why 36 times are not 36 independent units and how the rejected analysis could change both uncertainty and the question answered.

Video + follow-along: `u7_lesson4_live.html` (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.